



FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SAD DRAFT

SECTION 1

P1&P2

Group: Tech EcoPioneer

LECTURER: DR. MUHAMMAD IQBAL TARIQ BIN IDRIS

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1.0 Introduction

Koperasi Kakitangan KADA Kelantan Berhad previously relied on manual processes for its credit union services, which included member registration, loan applications, and financial transactions. Employees had to fill out physical forms to join the credit union and apply for loans, while clerks manually verified and entered these details into the system. The monthly contributions were managed through traditional banking methods, which caused the process to consume a lot of time and prone to errors.

To address these inefficiencies, our team has planned to develop an all-in-one online platform that can digitise and streamline the credit union services for Koperasi Kakitangan KADA Kelantan Berhad. With the platform, employees of the Kemubu Agricultural Development Authority (KADA) will be able to register as a member of the credit union online. The system will automatically verify their details, and will be sending to the clerk for approval which simplifies the registration process. The platform will allow members to apply for loans online. It will enable a faster and efficient review and approval by the Board of Directors. Also, monthly contributions can be made online in the platform via PayNet to ensure secure and timely transactions. Additionally, the members will have access to real-time information on the interest rates based on their contribution, loan maturity dates, and their financial status within the credit union.

The all-in-one online platform will help Koperasi Kakitangan KADA Kelantan Berhad to improve their operational efficiency and reduce their administrative burdens. By eliminating the manual paperwork and integrating all the functions into a single digital platform, we aim to increase members' satisfaction and engagement, and consequently supporting the financial well-being of KADA's employees.

2.0 Information gathering process

2.1 Method used

In order to plan and develop the super app for the Koperasi Kakitangan KADA Kelantan Berhad, our team gathered information through several methods. The process took place during an online meeting with Koperasi Kakitangan KADA Kelantan Berhad on Industry Day, 15 May 2024.

First, we held an online meeting with Koperasi KADA representatives, including the person in charge of the credit union operations. This meeting was impotent for us to discuss the current processes, challenges, and specific needs for the new superapp. The meeting gave us a clear understanding of how Koperasi KADA currently works and what improvements are needed.

During the meeting, we conducted interviews with the person in charge. We asked detailed questions about the current manual processes, such as how KADA employees register for the credit union, apply for loans, make monthly contributions, and how their contributions help them. These interviews helped us understand the pain points and areas where the new app could make a difference.

Additionally, we used WhatsApp for follow-up questions and clarifications. This allowed us to get quick responses and clear up any uncertainties from the meeting. We have collected comprehensive and accurate information. The data are essential for creating a super app that will streamline and digitise the credit union's operations, and make Koperasi Kakitangan KADA Kelantan Berhad more efficient and user-friendly.

2.2 Summary from method used

During the meeting, our team gathered important information from Koperasi Kakitangan KADA Kelantan Berhad. Here are the key questions asked and the responses received:

1. Possibility of Opening Koperasi Kakitangan KADA to the Public

Question: Is there a possibility for KADA to open Koperasi Kakitangan KADA to the public?

Answer: No. It is exclusive for KADA employees only.

2. Digitising Meetings with Board of Directors

Question: Regarding the Koperasi KADA meetings with the Board of Directors, do you wish to keep the manual system, or are you open to digitising the process?

Answer: Yes, we prefer to remain the manual system.

3. Monthly Salary Deduction for Share Capital

Question: Can the RM300 shares be paid via monthly salary deduction of RM55?

Answer: Yes, it can be deducted from salary at RM50 for six months, plus a RM5 contribution to the fund.

4. Share Capital

Question: What is the share capital and what is the maximum share?

Answer: The minimum share capital of RM300 is required to qualify for applying loans and is mandated by the Malaysia Co-operative Societies Commission of Malaysia (SKM). The maximum share capital is RM10,000.

5. Tabung Anggota, Simpanan Anggota and Simpanan Tetap

Question: What are the differences between Tabung Anggota, Simpanan Anggota and Simpanan Tetap?

Answer: The Tabung Simpanan Anggota is for members and their family welfare, such as death benefits, haji, and umrah. Simpanan Tetap are for members' retirement.

3.0 Background Study



Koperasi Kakitangan KADA Kelantan Berhad is a corporation that established on 29 August 1981 to serve the employees of the Kemubu Agricultural Development Authority (KADA). KADA itself is a state agency focused on agricultural development, aiming to improve agricultural productivity and the socio-economic status of farmers in Kelantan. The purpose of the corporation was to improve the economic and social well-being of its members through various services and initiatives including providing financial services, such as loans and savings in order to improve the quality of life of the members.

The cooperative is governed by Wan Badri b. Wan Omar, the Chairman of Koperasi Kakitangan KADA Kelantan Berhad. He is responsible for setting strategic directions and overseeing the management team, which handles day-to-day operations. The management team includes departments focused on different areas such as finance, member services, loans, retail, and investment.

The services provided by the corporation are saving and loans, which include saving accounts, personal loans and emergency loans such as Al Bai Loan, Al Innah Loan, Al Quardhul Hasan Loan and much more. Besides that, investment such as fixed deposits, property investment and equity investment can also be provided by the corporation to the members.

4.0 Problem Statement

During the industry visit, we found that there are several issues occurring on the current system of Koperasi Kakitangan KADA Kelantan Berhad. The issues are listed in below:

- 1) Inefficient record-storing method
 - Most records, including registration and records of applying for a loan, are recorded and stored on paper.
 - Inefficient for the staff when they need to review or look for certain records.
 - May easily cause loss of information
- 2) Poor Communication Systems
 - The approval for loan of the members requires higher-up to undergo face-to-face meeting
 - May cause waste of time and energy
- 3) Slow approval process
 - Processes such as loan approvals, member applications, and financial transactions often require multiple layers of manual approval
 - May leads to delay of results delivery

5.0 Proposed Solutions (include feasibility study – technical, operational, economical - CBA)

To address the challenges posed by inefficient record-keeping methods, **implementing a digital record-keeping system emerges as a viable solution.** This system would transition the organization away from paper-based records, opting instead for cloud-based storage and database management systems. Through document scanning and digitization tools, existing paper records could be seamlessly integrated into the digital infrastructure. Staff training would be essential to ensure a smooth transition, educating employees on the new procedures and tools for digital record-keeping. Additionally, the system would enforce access controls and permissions to safeguard sensitive data, while robust backup and recovery protocols would mitigate the risk of data loss.

Furthermore, the introduction of an online approval system offers a solution to the inefficiencies plaguing the organization's approval processes. **By selecting and customizing an online approval software solution, the organization can streamline loan approvals and transaction processes.** Integration with existing systems would ensure seamless operation, while stringent security measures such as encryption and authentication would safeguard sensitive data. Operational benefits would include the automation of manual approval workflows, reducing

processing delays and improving overall efficiency. Staff training on the new system would be crucial for successful adoption, supplemented by monitoring and reporting tools for oversight.

Moreover, **upgrading communication systems with an enhanced online platform** can address the shortcomings of the organization's current communication channels. By selecting a suitable online communication platform equipped with features like video conferencing and notifications, the organization can facilitate virtual meetings and real-time communication. Technical considerations would include ensuring compatibility across devices and implementing robust security measures to protect sensitive information. Operational implementation would involve staff training on platform usage and integration with existing communication channels. The platform's notification capabilities would enhance responsiveness, alerting stakeholders to important updates and events in real-time, thus improving overall communication efficiency.

Technical feasibility

1. Digital Record-Keeping System:

- Compatibility: Ensure it works with existing hardware and software.
- Integration: Seamlessly connect with current systems and databases to prevent data management disruptions.

2. Online Approval System:

- Accessibility: Ensure usability across all devices and browsers.
- Integration with External Services: Smooth integration with services like courier partners, with thorough testing to ensure functionality.

3. Enhanced Communication Platform:

- Support: Operate on various operating systems and devices to meet diverse user preferences.
- Security: Implement robust security measures, including encryption protocols, to protect sensitive data during communication.

Operational Feasibility

1. Digital Record-Keeping System:

- User Acceptance: Provide tailored training to ensure users are proficient.
- Training Programs: Address diverse needs of different user groups.

2. Online Approval System:

- Integration: Seamlessly fit into existing approval workflows.
- Training and Support: Provide adequate training to minimize disruptions.

3. Enhanced Communication Platform:

- Usability: Conduct usability testing and provide training.
- Collaboration: Work with IT and end-users to identify and fix issues.

6.0 Objectives

1. Identify problems and opportunities in KADA's current manual processes
2. Determine human information requirements from KADA employees for system functionalities
3. Analyse system needs for Koperasi KADA daily operation
4. Design the recommended system to Koperasi KADA's needs.
5. Develop and document the software for Koperasi KADA
6. Test the system thoroughly and establish a maintenance plan
7. Implement the system, collect feedback, and perform evaluations

7.0 Scope of the Project

The scope of this project includes developing a system to digitise and streamline the credit union services for Koperasi Kakitangan KADA Kelantan Berhad. Besides, the system will be designed solely for desktop use and will not be optimised for mobile devices, restricting its usability to desktop platforms only. Furthermore, the system will be exclusive to KADA employees and will not be opened to the public. The system will support only four entities which are new users, members, staff and board of directors. Moreover, the system will not cover any additional functionalities beyond new member registration, loan applications, saving applications and report summarization.

8.0 Project Planning

8.1 Human Resource

Our group consists of 5 members which are Ng Yu Hin, Chew Chiu Xian, Lau Yan Kai, Evelyn Goh Yuan Qi and Dheshieghan A/L Saravana Moorthy. Each person has been assigned to a specific role as listed below:

Project Manager: NG YU HIN

- Oversee the entire project
- Manages tasks and progress of such tasks
- Assigns tasks to group members

Analyst: DHESHIEGHAN A/L SARAVANA MOORTHY & LAU YAN KAI

- Collect data from various sources, conduct interviews, surveys, and analyse current systems
- Analyse data to identify insights, conduct feasibility studies, and evaluate risks.

Designer: CHEW CHIU XIAN

- Prepare rough design for website wireframe
- Refine design for website prototype

Advisor: EVELYN GOH YUAN QI

- Reviews and monitors progress
- Recommend improvements
- Assist in problem-solving

8.2 Work Breakdown Structure (WBS)

Activity	Task	Predecessor	Duration (Days)
Phase 1: Project Proposal and Planning			10
A	1.1 Interview User Management	None	1
B	1.2 Introduction	A	1
C	1.3 Background Study	B	1
D	1.4 Problem Statement	C	1
E	1.5 Proposed Solutions 1.5.1 Technical Feasibility 1.5.2 Operational Feasibility 1.5.3 Economical Feasibility	D	1
F	1.6 Objectives	E	1
G	1.7 Scope of the project	F	1
H	1.8 Project Planning 1.7.1 Human resource 1.7.2 Work Breakdown Structure 1.7.3 PERT Chart 1.7.4 Gantt Chart	G	2
I	1.9 Benefit and overall summary of proposed system	H	1
Phase 2: Information System Gathering and Requirement			9
J	2.1 Information gathering process 2.1.1 Method used 2.1.2 Summary from method used	I	3
K	2.2 Requirement Analysis (AS-IS Analysis) 2.2.1 Current business process 2.2.2 Functional Requirement 2.2.3 Non-functional Requirement 2.2.4 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)	J	6
Phase 3: System Analysis & Design			18
L	3.1 DFD TO-BE	K	9
M	3.2 Design Interface	L	9

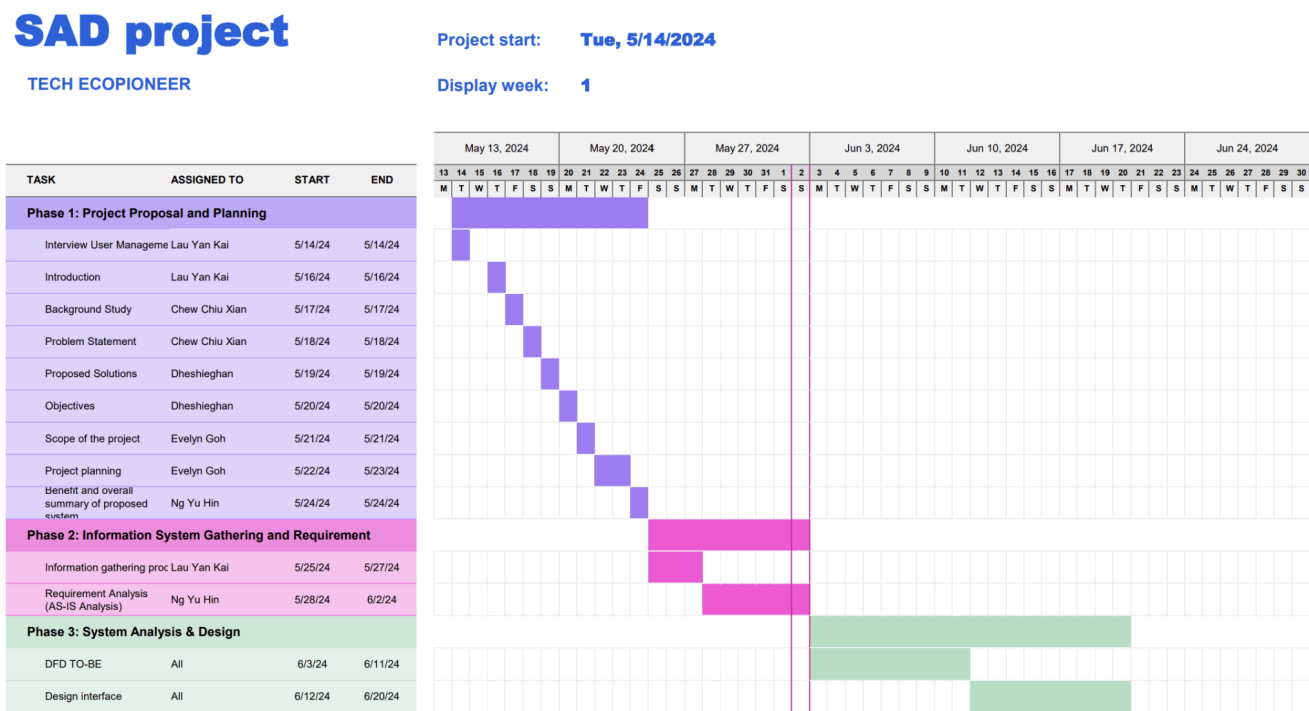
8.3 PERT Chart (based on WBS)



Note: Assume all durations are in days

Path 1: A-B-C-D-E-F-G-H-I-J-K-L-M (37 days)

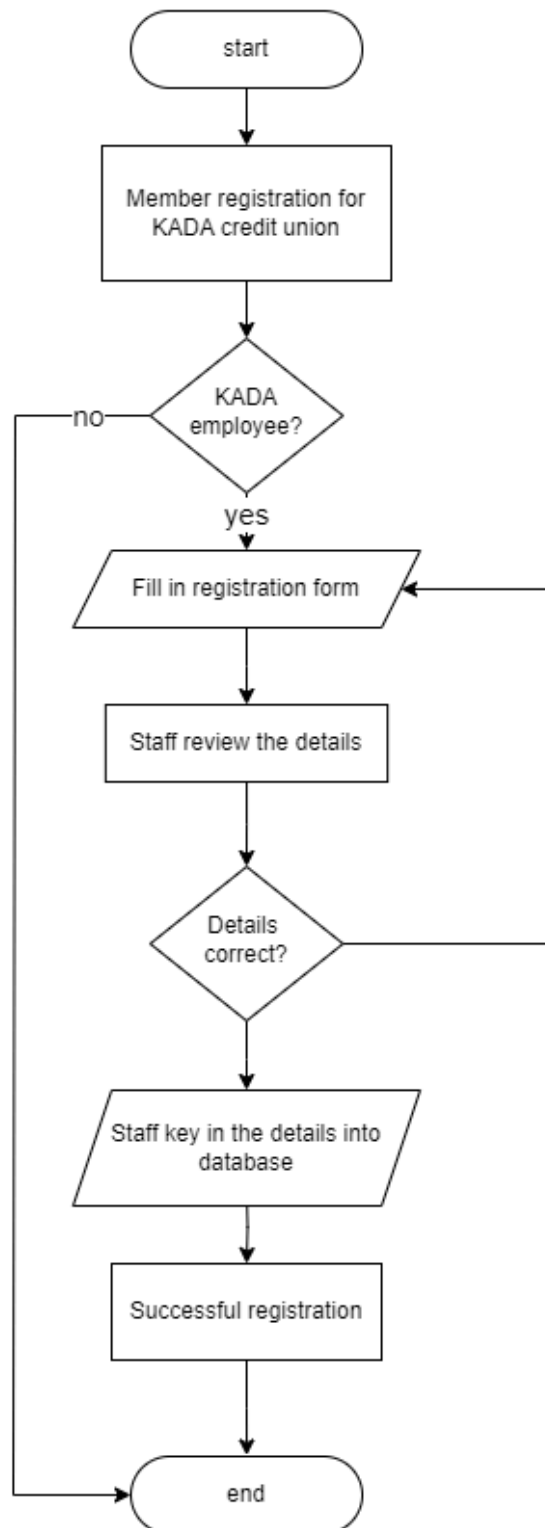
8.4 Gantt Chart



9.0 Requirement Analysis (based on AS-IS analysis)

9.1 Current business process (scenarios, workflow)

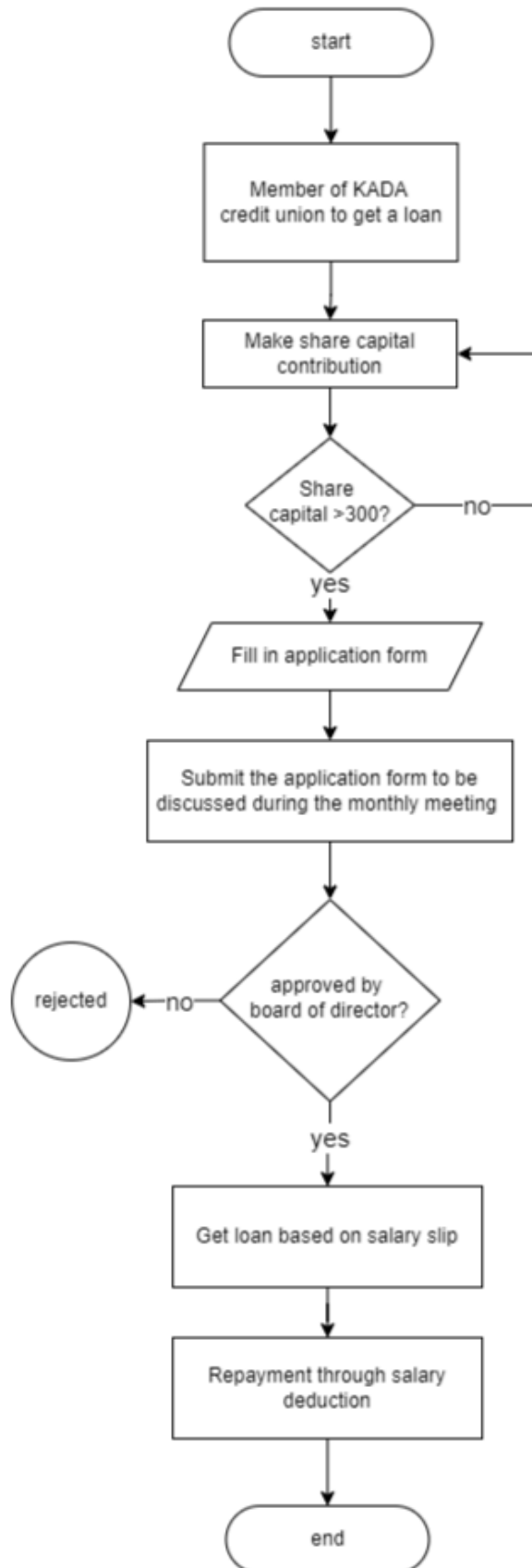
Scenario 1: *In-Person Member Registration through Form*



Scenario 1 workflow:

1. Is the applicant employee of KADA?
 - 1.1. If yes, proceed to the next step.
 - 1.2. If no, the application is rejected. End.
2. Fill in the application form
3. Staff review the application form
4. Is the application form details correct?
 - 4.1. If yes, proceed to the next step.
 - 4.2. If no, pass back the application form, and ask the applicant to resubmit.
5. Staff key in the details of the member in system
6. Successful registration
7. End

Scenario 2: In-Person Loan Application through Form



Scenario 2 workflow:

1. Make a share capital contribution
2. Is the share capital greater than or equal to 300?
 - 2.1. If yes, proceed to the next step.
 - 2.2. If no, go back to step 1.
3. Fill in the application form
4. Submit the application form to be discussed during the monthly meeting of the board of directors
5. Is the application approved by the board of directors?
 - 5.1. If yes, proceed to the next step.
 - 5.2. If no, the application is rejected. The applicant could make an application again.
6. Obtain a loan based on the salary slip
7. Repay the loan through salary deduction
8. End

9.2 Functional Requirement (input, process and output)

Functional Requirements	Input	Process	Output
1. User/New Member Registration	Personal info and details	Submission of form	Acknowledgement email/message (indicate submission was successful)
2. Data gathering	Information from forms	Collection of the information	Data gathered ready for verification
3. Data verification	Submitted forms and supporting documents	Verification by staff	Verified data or flagged issues for follow up
4. Data entry	Verified data	Manual/Automate entry from forms	Data stored in database
5. Approval process	Verified registration/request	Review by management	Registration/Request status updated
6. Notification	Registration/Request status	Create email/message	Sent notification to user/new member
7. Authorization management	Approved registration details	Identification assignment	Provide access credentials
8. Report	Registration data	Data summarization	Registration report

9.3 Non-functional Requirement (performance and control)

Non-functional Requirement	Requirement	Details
Performance	Process and store registration/requests promptly	Should handle numerous registration/requests
	Quick response time	Increase user satisfaction/Reduce frustration
	High system throughput	To support a high quantity of concurrent users
Control	Manage access control	Restrict certain access to sensitive data & functionalities based on user roles
	Implement data encryption	Apply strong encryption algorithms
	Apply backup and recovery	Perform daily backups and ability to recover progress within system

9.4 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)

Context Diagram

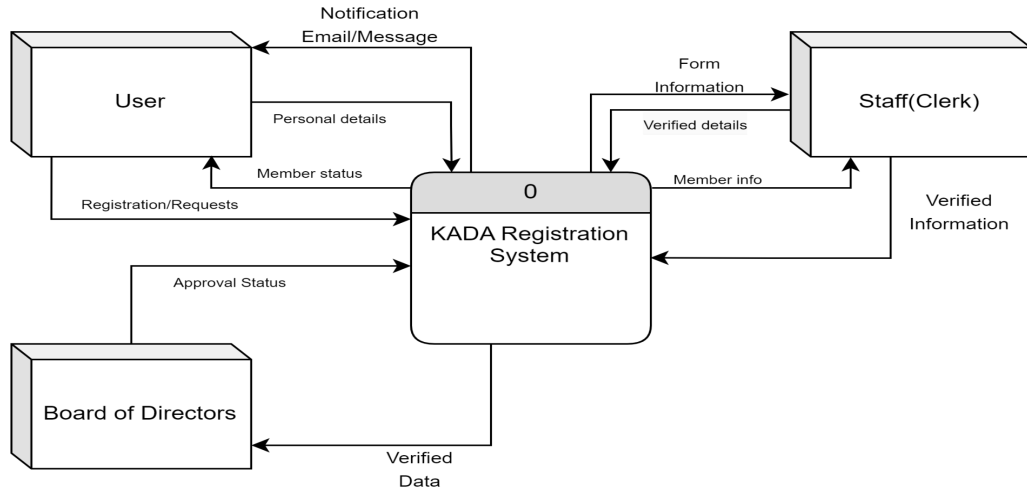
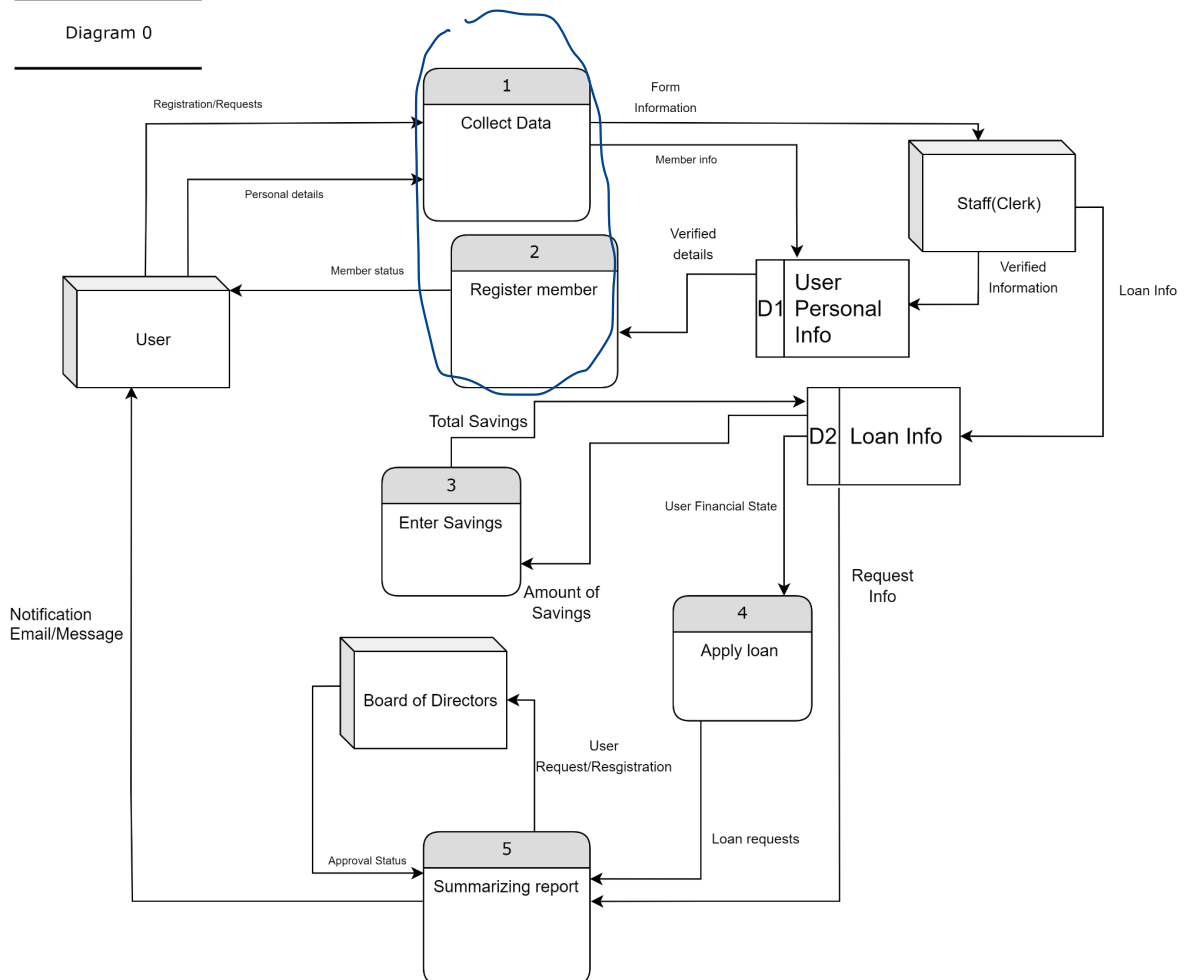
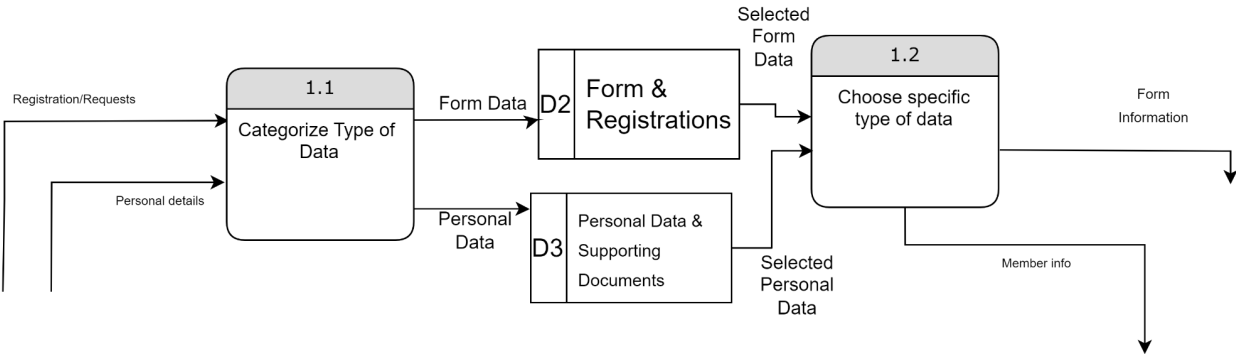


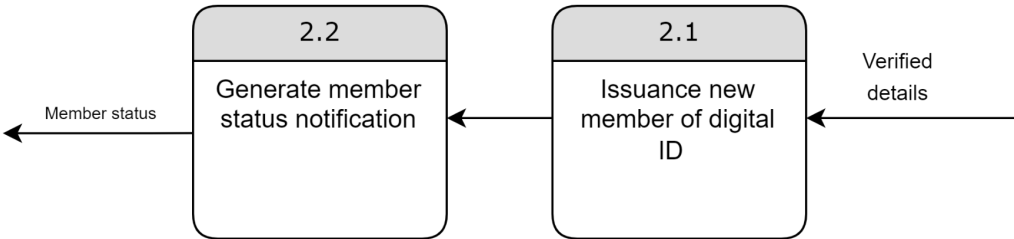
Diagram 0



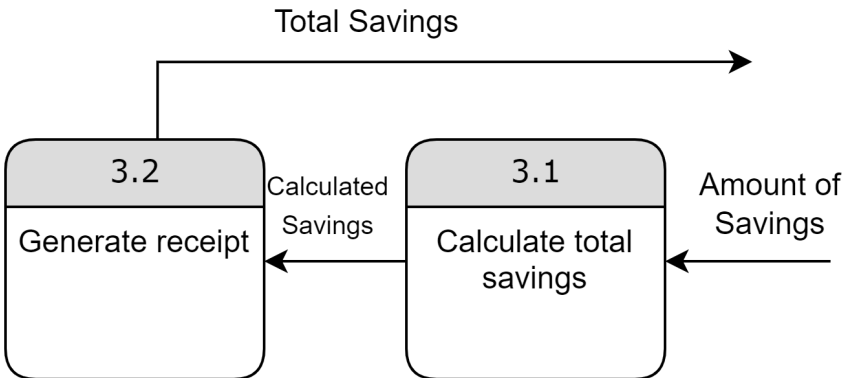
Child Diagram 1

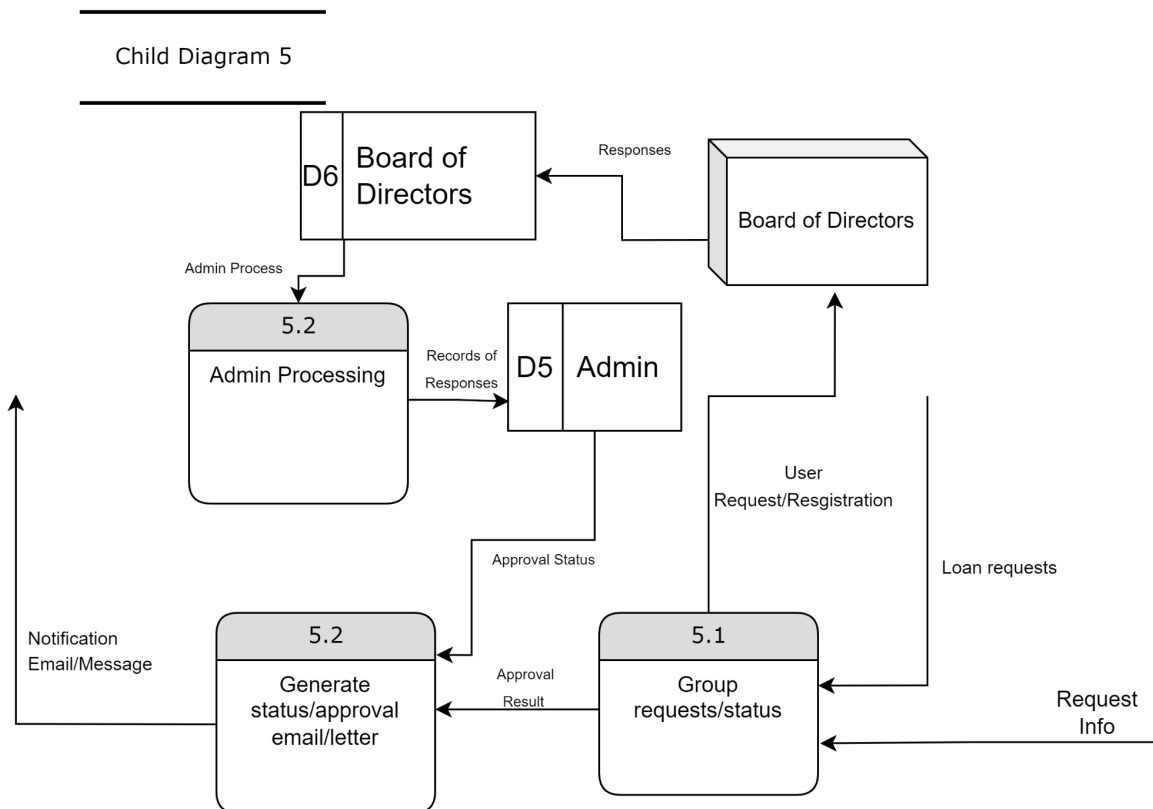
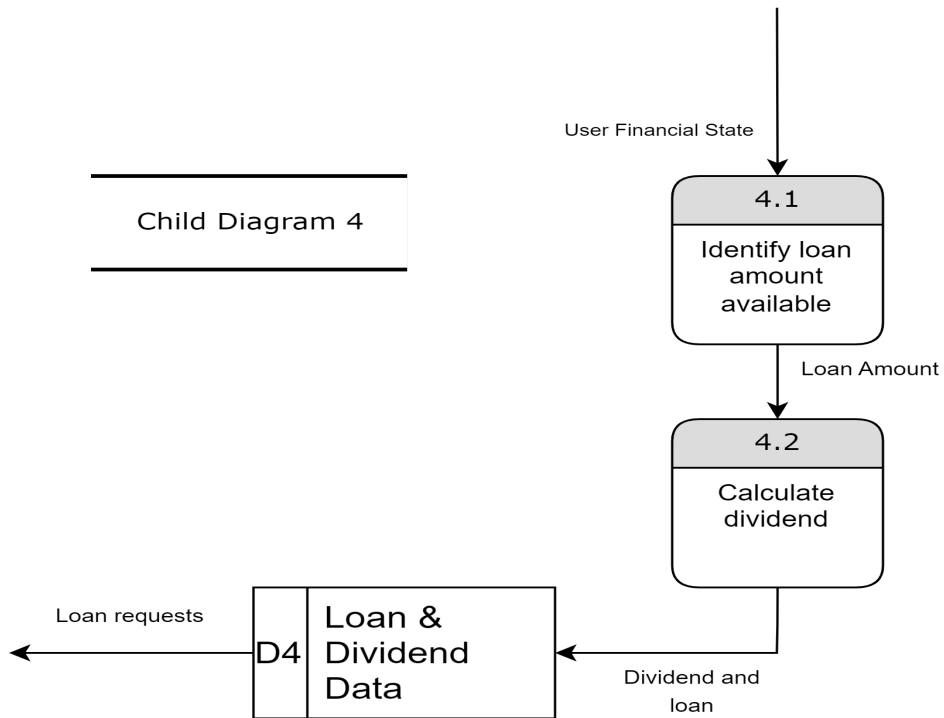


Child Diagram 2



Child Diagram 3





10.0 Benefit and Overall Summary of Proposed System

The AS-IS registration system of Lembaga Kemajuan Pertanian Kemubu has several parts that are done manually.

These can be changed to automate so that it is more efficient and minimises human mistake.

Some facilities also can be added to the system to decrease the workload.

The function of the To-Be system are as follows:

1. Automated registration
2. Online Savings
3. Online apply loans
4. Automated dividend and savings calculation
5. Automated generated notifications
6. Database for user information

Snapshot for Project Status

 Status updates

Add update

 lauyankai updated 7 minutes ago ...

 Start date:

2024-05-23

 Target date:

2024-06-01

 Status:

On track

Completed project 1 and project 2.

Snapshot for Backlog of Project

  lauyankai / Projects / all-in-one online platform for Koperasi Kakitangan KADA Kelantan Berhad

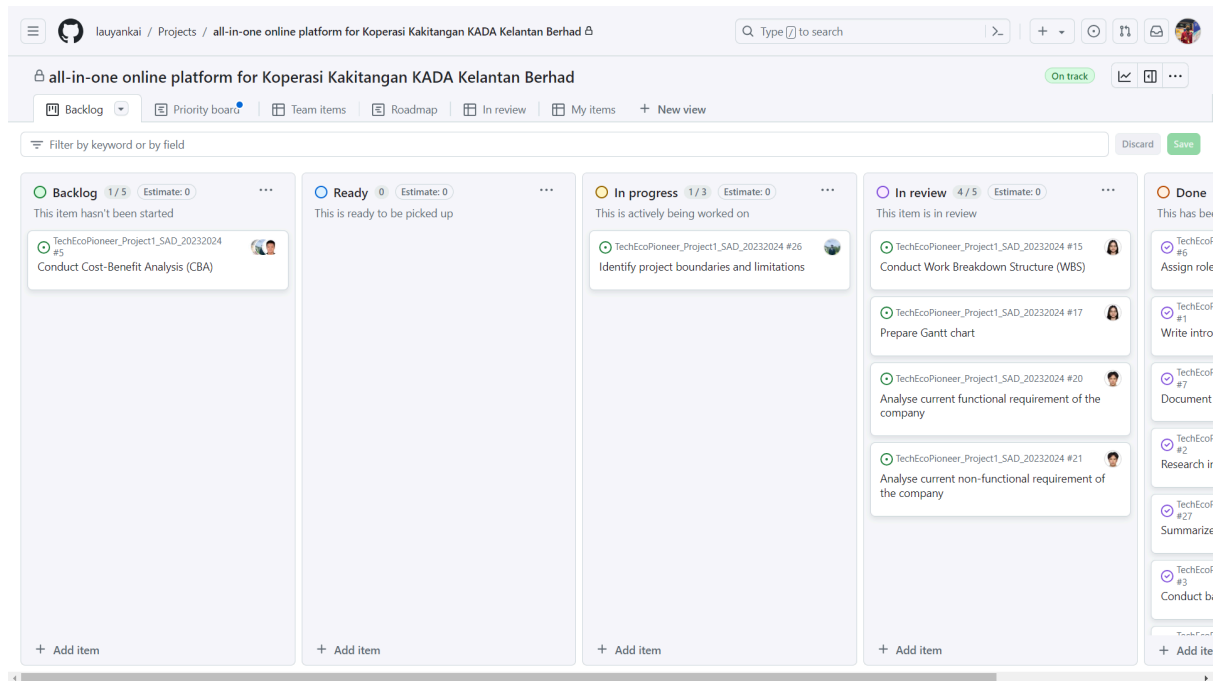
 Backlog  Priority board  Team items  Roadmap  In review  My items  New view

Discard Save

	Title	Assignees	Status	Priority	Estimate
1	 Assign roles and responsibilities #6	 lauyankai	Done		
2	 Write introduction section #1	 lauyankai	Done		
3	 Document methods used #7	 lauyankai	Done		
4	 Research information gathering methods #2	 lauyankai	Done		
5	 Summarize findings #27	 lauyankai	Done		
6	 Conduct background study #3	 cxchew	Done		
7	 Document findings from background study #8	 cxchew	Done		
8	 Identify the core problem facing #4	 cxchew, dheshiegh...	Done		
9	 Document problem statement #9	 cxchew	Done		
10	 Conduct Cost-Benefit Analysis (CBA) #5	 cxchew, dheshiegh...	Backlog		
11	 Identify proposed solution #10	 dheshieghan227	Done		
12	 List project objectives #11	 dheshieghan227	Done		
13	 Identify project boundaries and limitations #26	 dheshieghan227	In progress		
14	 Document objectives #12	 dheshieghan227	Done		
15	 Document scope #13	 evelynngoh	Done		

 You can use `Control + Space` to add an item

Snapshot for Kanban Board Integration



Snapshot of Project Roadmap

